Roll No: | | | | | | | | | | | | | |

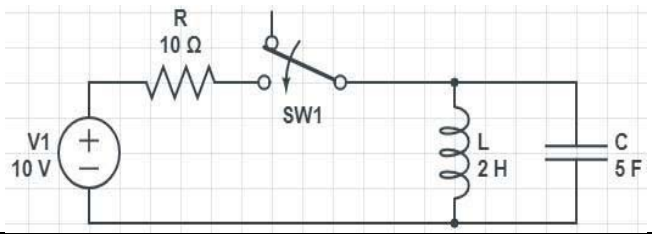
**BTECH**
**(SEM III) THEORY EXAMINATION 2021-22**
**SIGNALS & SYSTEMS**

***Time: 3 Hours*** ***Total Marks: 70***

**Notes:**

- *Attempt all Sections and Assume any missing data.*
- *Appropriate marks are allotted to each question, answer accordingly.*

| **SECTION-A** | Attempt **All** of the following Questions in brief | Marks **(7X2=14)** |
|---|---|---|
| Q1(a) | Explain how aperiodic signals can be represented by Fourier transform. | |
| Q1(b) | What are the major classifications of the signal | |
| Q1(c) | What are the Conditions for a System to be LTI System? | |
| Q1(d) | Find the Fourier transform of x(t)=sin(wt) | |
| Q1(e) | Define time invariant and time varying systems. | |
| Q1(f) | What are the properties of convolution? | |
| Q1(g) | What is the use of Laplace transform | |

| **SECTION-B** | Attempt **ANY THREE** of the following Questions | Marks **(3X7=21)** |
|---|---|---|
| Q2(a) | (i) What is the periodicity of $x(t) = e^{j100\Pi t} + 30^o$<br>(ii) Find the fundamental period of the signal $x(n) = 3\, e^{j3\Pi(n+1/2)}$<br>(iii) Is the system describe by the equation y(t) = x(2t) Time invariant or not? Why<br>(iv) What is the period T of the signal x(t) = 2cos (n/4)? | |
| Q2(b) | Find the Fourier coefficients of $x(t)=1+\sin 2\omega t+2\cos 2\omega t+\cos[3\omega t+\pi/3)$. | |
| Q2(c) | Explain Modelling of mechanical system and electrical system and then give the analogy between electrical and mechanical system. | |
| Q2(d) | State and prove convolution property of Laplace transform and then using this property find Laplace transform of $\frac{s}{(s+1)(s+2)}$ | |
| Q2(e) | For the circuit given below, Initially switch is closed for a long time and steady state condition has reached. At t =0 switch is opened. Find the expression of current through inductor   | |

| **SECTION-C** | Attempt **ANY ONE** following Question | Marks **(1X7=7)** |
|---|---|---|
| Q3(a) | State and prove time shifting and differentiation properties of Z transform. | |
| Q3(b) | A signal has Laplace transform $X(s) = \frac{(s+2)}{(s^2+4s+5)}$<br>Find the Laplace transform Y(s), of the following signals<br>(i) $y(t) = t\,x(t)$ (ii) $y(t) = e^{-t}x(t)$ | |

Source: AKTU official examination paper - www.aktunotes.in

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| **SECTION-C** | Attempt **ANY ONE** following Question | Marks **(1X7=7)** |
|---|---|---|
| Q4(a) | Obtain the state model for the given transfer function<br>$T(s) = Y(s) / U(s) = K (b_1s+b_2) / (s^3+a_1s^2+a_2s+a_3)$ | |
| Q4(b) | Obtain the state transition matrix of a system given by.<br>$A=\begin{bmatrix} -1/2 & -5/2 \\ 1/2 & -7/2 \end{bmatrix}$ | |

| **SECTION-C** | Attempt **ANY ONE** following Question | Marks **(1X7=7)** |
|---|---|---|
| Q5(a) | Find even and odd component of the following signals<br>(i) x(t)= cost + sint + cost sint<br>(ii) x(n) = {1,2,1,4,5,0,3} | |
| Q5(b) | State and prove the time delay theorem and Parsavel's theorem of Z-transform. | |

| **SECTION-C** | Attempt **ANY ONE** following Question | Marks **(1X7=7)** |
|---|---|---|
| Q6(a) | Obtain the trigonometric Fourier series for the half wave rectified sinewave. | |
| Q6(b) | State and prove duality property of Fourier transform. Find the inverse Fourier transform of,<br>$X(j\omega) = \begin{cases} 2cos\omega, & \lvert\omega\rvert < \pi \\ 0, & \lvert\omega\rvert > \pi \end{cases}$ | |

| **SECTION-C** | Attempt **ANY ONE** following Question | Marks **(1X7=7)** |
|---|---|---|
| Q7(a) | Find the inverse Laplace transform of given function by using convolution theorem<br>(i) $x(s)= 1/(s^2+a^2)^2$<br>(ii) $x(s)= s/(s+1)(s+2)$ | |
| Q7(b) | Explain state transition matrix, its physical significance and properties. | |